
National Physical AI Oncology Trial Platform

Document: program white paper
Scope: a national trial network
Premise: the patient is the party the verification gate protects

Kevin Kawchak , CEO ChemicalQDevice | 10.5281/zenodo.xxxxxxxx | June 5, 2026

Abstract.

This white paper template describes a national Physical AI oncology trial platform: its mission, architecture, governance, and roadmap, with verification before generation as the shared safety rule.

Disclaimer: This work is independent and not endorsed or sponsored by trial sponsors, FDA, CROs, sites, IRBs, regulators, or medical societies; and was generated using Artificial Intelligence.

Keywords

1 Mission

The platform’s mission is to let cancer trials adopt embodied AI at the speed modernization demands while holding patient safety first, through a single verify-before-generate rule applied network-wide.

2 Architecture

The platform federates sites under a shared verification gate and a common audit-trail standard, so a procedure cleared at one site is reconstructable at any other.

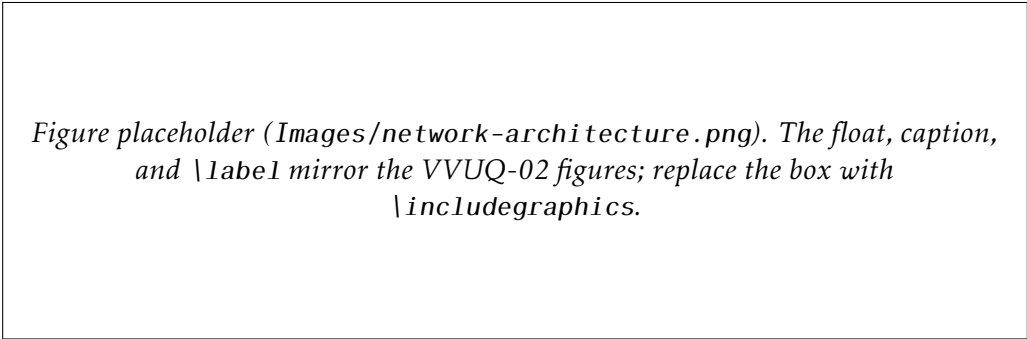


Figure 1: Figure 1. Placeholder network architecture across federated sites.

3 Governance and equity

Governance keeps the patient, not the vendor or payer, as the protected party, and binds nondiscrimination and accessibility duties into the gate criteria rather than leaving them to discretion.

4 Platform metrics

Layer	Function	Metric
site	gated procedures	fraction equals 1.0 reconstructable records
network	shared audit trail	
governance	human oversight	escalation honored

Table 1: Table 1. Platform layers and metrics (modeled on VVUQ-05 Table 1).

5 Roadmap

The platform matures in three stages.

1. **Pilot.** stand up the gate at a lead site
2. **Federate.** extend the shared audit standard across sites
3. **Scale.** onboard the national network under one rule

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Ethical Disclosures

The author declares no competing interests. This is an independent draft and is not endorsed, sponsored, or approved by any trial sponsor, CRO, site, IRB, regulator, or medical society, nor by CFR, ICH, or FDA. All supporting numbers are simulation or illustrative results.

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Cite This Article

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Data Availability

Supporting records are openly available in the [kevinkawchak/cancer-automated](#) and [kevinkawchak/physical-ai-oncology-trials](#) GitHub repositories and are archived on Zenodo. All data sets are synthetic or illustrative and reproducible from a deterministic seed; no patient-identifiable information is included.

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